

Application of artificial intelligence techniques in meat processing: A review

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Abstract

The field of meat processing plays a critical role in the food industry and has seen increasing adoption of artificial intelligence (AI) technology with rapid technological advancements. AI technology has tremendous potential for enhancing production efficiency and product quality in meat processing. However, further research and exploration are necessary to tackle the challenges posed by the use of AI technology. This article details the implementation of AI technology in meat processing, focusing on carcass classification, automation and intelligent processing, and meat-quality detection. We aim to provide inspiration to researchers and industry professionals and promote the advancement of AI technology in the meat processing sector.

Practical applications

Our review article showcases the potential industrial applications of artificial intelligence (AI) techniques in the meat processing industry. AI technology can greatly improve production efficiency and product quality in meat processing. By implementing AI algorithms, meat processors can accurately classify carcasses, automate various processing tasks, and detect meat quality with higher accuracy. These advancements can lead to increased profitability and improved food safety in the industry. We hope to provide valuable insights for researchers and industry professionals, encouraging them to further explore and adopt AI technology in the meat processing sector.

KEYWORDS

artificial intelligence technology, automation, carcass classification, meat, processing, quality inspection

1 | INTRODUCTION

Meat comprises proteins, vitamins, dietary fats, and other essential nutrients. It is a crucial and fundamental component of human beings' daily diet (Grujić, 2010). China's domestic poultry, pork, beef, and lamb production amounted to 750,000 tons. China has now surpassed other countries to become the world's biggest meat consumer (Wang, 2022). However, meat is a crucial protein source in human diets, but its consumption could pose a considerable health hazard without appropriate safety measures. Contaminants such as bacteria, parasites, and heavy metals must be managed appropriately to prevent food poisoning and related health concerns (Adesokan &

Raji, 2014). Raw meat is relatively uniform in both form and taste, and its inherent instability makes it unsuitable for prolonged storage or catering to the diverse needs of consumers. Processing provides a means to address these challenges.

Meat processing, comprising slaughtering, cutting, packaging, storage, and transportation, is a crucial aspect of the food industry. On the one hand, it can convert raw meat into safer, more readily preserved products to cater to consumer needs via procedures like cutting, smoking, curing, dehydrating, fermenting, and freezing. On the other hand, the processing of meat constitutes a crucial responsibility for meat processing companies as they strive to advance processing efficiency and allocate production resources to enhance their economic efficiency.

Traditional meat processing methods involve procedures such as manual slaughtering, manual boning, and manual carcass grading, among others. Despite their effectiveness, these methods necessitate significant manual involvement, rendering the process time-consuming and less efficient. Additionally, manual methods can cause meat damage, waste, unevenness, and other issues (De La Iglesia et al., 2020). With the advancement of technology, various automated techniques, including mechanical cutting and boning, have been introduced in meat processing. These techniques have the potential to improve both the quality and yield of the final product, as well as enhance processing efficiency. Nevertheless, they pose challenges in handling carcasses of varying shapes or adapting to different environments (Seaton, 2022).

Changing consumer demands and rising global competition are driving meat companies to adopt novel processing technologies (Raj et al., 2013). Thanks to significant advances in data management, algorithms, and computational capability, artificial intelligence (AI) technology is developing rapidly and is finding applications within various domains, including medicine, geology, and beyond (Liu et al., 2021; Mulvenna et al., 1991). The incorporation of AI technology in meat processing allows for automatic identification, detection, and classification tasks to be completed, resulting in an intelligent production line. It effectively remedies the limitations of traditional processing and boasts substantial potential for development in the meat processing industry.

This article presents an overview of the utilization of AI technology in meat processing, examining the technical principles and practical applications. It aimed to provide a useful reference for further explorations in meat processing.

2 | OVERVIEW OF AI TECHNOLOGIES

AI is a highly comprehensive interdisciplinary discipline developed based on multiple disciplines such as computer science, cybernetics, information theory, and neuropsychology. AI technology, together with space technology and atomic energy technology, is regarded as one of the three major scientific and technological achievements of the 20th century, expanding the functions of the human brain and realizing the automation of mental work to a certain extent.

AI is the study of the research and design of intelligent subjects, which are systems that can observe their surroundings and take action to achieve their goals (Russell & Norvig, 2004). AI technology is the technology of intelligence realized in machines by artificial means—that is, the technology of constructing an intelligent machine or an intelligent system through computer programs and algorithms so that it can simulate, extend, and expand human intelligence. It can learn from data, and perform self-optimization and autonomous decision-making, to realize the cognition, reasoning, learning, and decision-making capabilities that human intelligence possesses.

At present, with the development of intelligent science and technology and the wide application of computer network technology, AI technology is applied to more and more fields, including machine

learning, expert systems, computer vision system (CVS), natural language processing, etc. The following is a brief introduction to several major research areas.

2.1 | Knowledge representation and reasoning

Knowledge representation and reasoning is an important area of AI, where knowledge is represented in a form suitable for computer processing, and these representations are used for reasoning and inference.

Common knowledge representation methods include first-order predicate logic representation, generative representation, frame representation and knowledge map, semantic network, and so on. The reasonable representation of knowledge lays the foundation for subsequent reasoning methods and expert systems. Reasoning method is an important research topic of AI, is an important method of solving problems, can make the computer in possession of knowledge, and also can think, that is, can apply knowledge to solve problems. At present, people have proposed a variety of reasoning methods that can be realized on the computer, which can be roughly divided into deterministic reasoning methods and uncertainty reasoning methods. Figure 1 shows the definitions and general content of these two types of reasoning methods.

An application that combines fuzzy reasoning and neural networks is the adaptive neuro-fuzzy inference system (ANFIS). The ANFIS is not only able to represent fuzzy rules in knowledge, but it can also reason and adapt using neural networks. It combines the advantages of fuzzy logic and neural networks to express uncertain knowledge and adaptive learning and reasoning based on data.

2.2 | Machine learning

Machine learning enables computers to simulate human learning behavior, automatically acquire knowledge and skills through learning, continuously improve performance, and realize self-improvement, which is an important research field of AI technology. Machine learning mainly studies the three aspects of learning mechanism, learning method, and learning system, and can achieve the internal behavior of acquiring knowledge, accumulating experience, and discovering laws, and the external performance of improving performance, adapting to the environment, and realizing system self-improvement. The goal is to build algorithms that can automatically learn and improve to solve specific tasks or problems. In machine learning, algorithms are trained on data sets and use patterns and associations in the data to make intelligent predictions and decisions.

Machine learning can be categorized from different perspectives and according to different methods. The most common classification method currently used is to classify the system according to how it learns, machine learning is categorized into supervised learning, unsupervised learning, and reinforcement learning. Figure 2 shows the classification of machine-learning and some common algorithms.

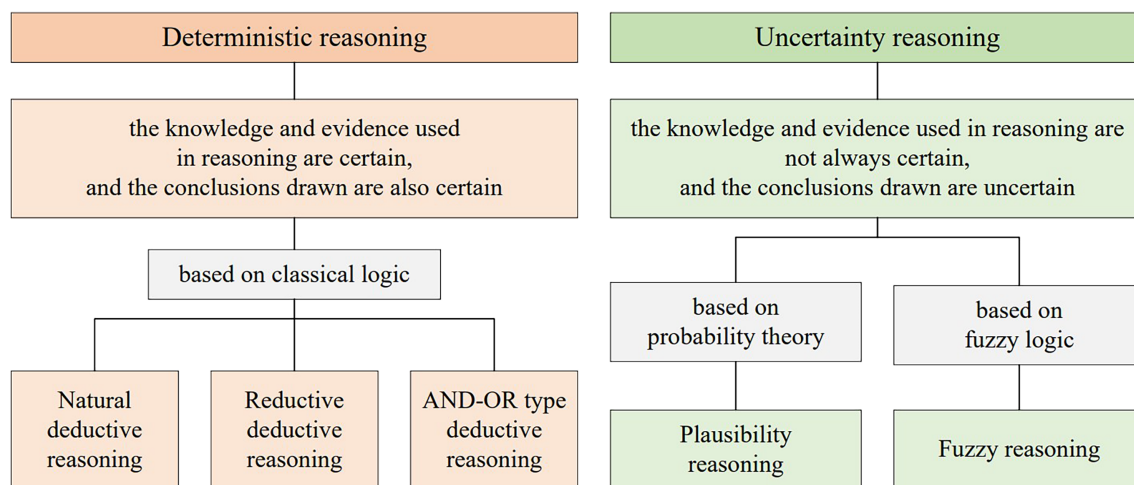


FIGURE 1 Examples of deterministic and uncertain reasoning methods.

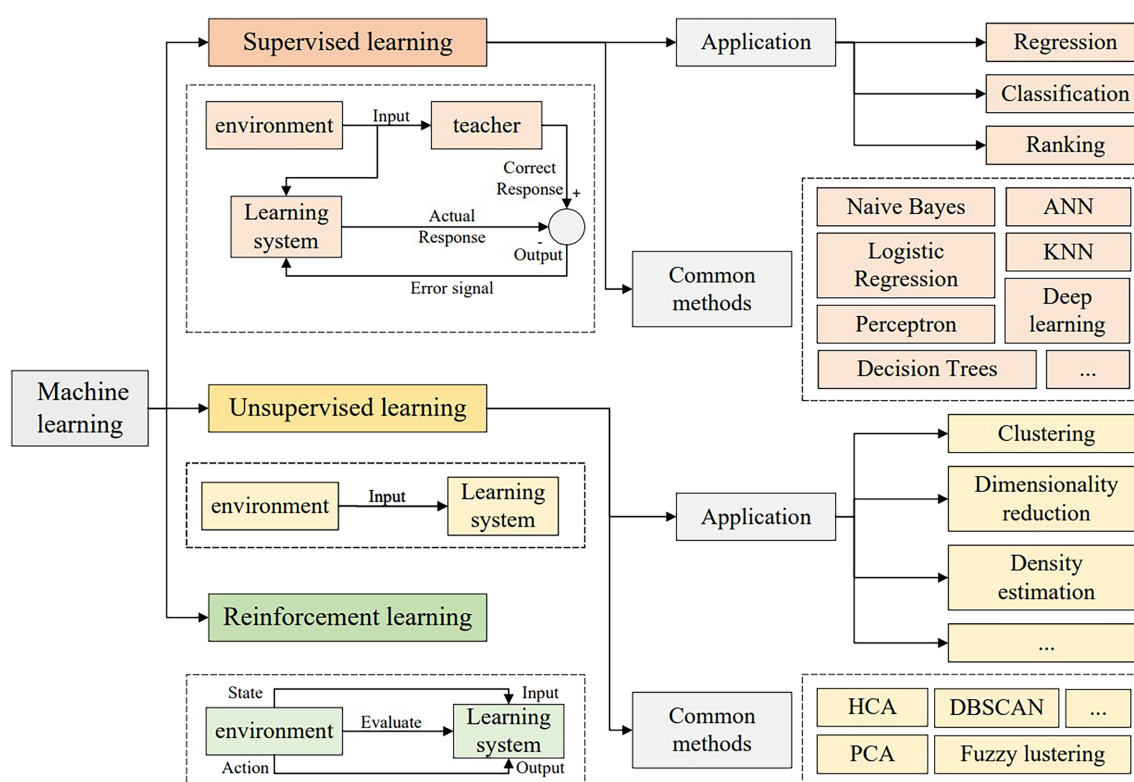


FIGURE 2 Machine-learning classification chart. ANN, artificial neural network; DBSCAN, density-based spatial clustering of applications with noise; HCA, hierarchical cluster analysis; KNN, K-means clustering algorithm; PCA, principal component analysis.

Intelligent optimization algorithms are also often considered to be part of machine learning. Intelligent optimization algorithms are problem-solving algorithms designed by researchers inspired by and imitating the laws of nature and biology, such as heredity, including genetic algorithms, simulated annealing algorithms, forbidden search algorithms, immune algorithms, etc., and usually include two major categories of methods: evolutionary computation and swarm intelligence.

2.3 | Computer vision

CVS is an important branch of AI, the study of how computers acquire, process, analyze, and understand digital images, video, and other visual input, to give computers the ability to mimic the human visual system.

CVS trains machines to rely on cameras, data, and algorithms to perform these functions and to ensure that the work is completed in

the shortest possible time. While CVS technology differs from human reliance on biological structures such as the retina for recognition and analysis, it draws on techniques and methods from the fields of mathematics and statistics, machine learning and deep learning, image processing and computer graphics, sensor technology, databases, and big data (Kruger et al., 2013; Medathati et al., 2016). Therefore, CVS technology recognition speed, and high precision, in many areas of CVS recognition speed and ability have even exceeded humans.

Common CVS tasks include image classification, target detection and recognition, video analysis, behavior recognition, etc., and the implementation of these tasks can be seen as the process of constructing models using geometry, physics, and statistical learning theory and refining symbolic information from visual data. In this process, the most critical is to extract symbolic information, to convert

the external input light signal into the understanding and cognition of the outside world, in other words, to extract the features of the image. Figure 3 describes two directions of feature extraction in the field of CVS and shows the general flow of CVS tasks.

In food-related fields, early tasks are often implemented using some simple image processing methods to extract features such as color, texture, or shape. For instance, after employing image segmentation methods to locate the region containing the food, features like color and texture can be extracted and fused for dish image recognition (Zhu et al., 2011). Considering the complementarity between different hand-crafted features, more methods are fused to combine different hand-crafted features to improve the model performance, such as some studies have focused on fusing various types of image features, including Scale-invariant Feature Transform (SIFT), Gabor

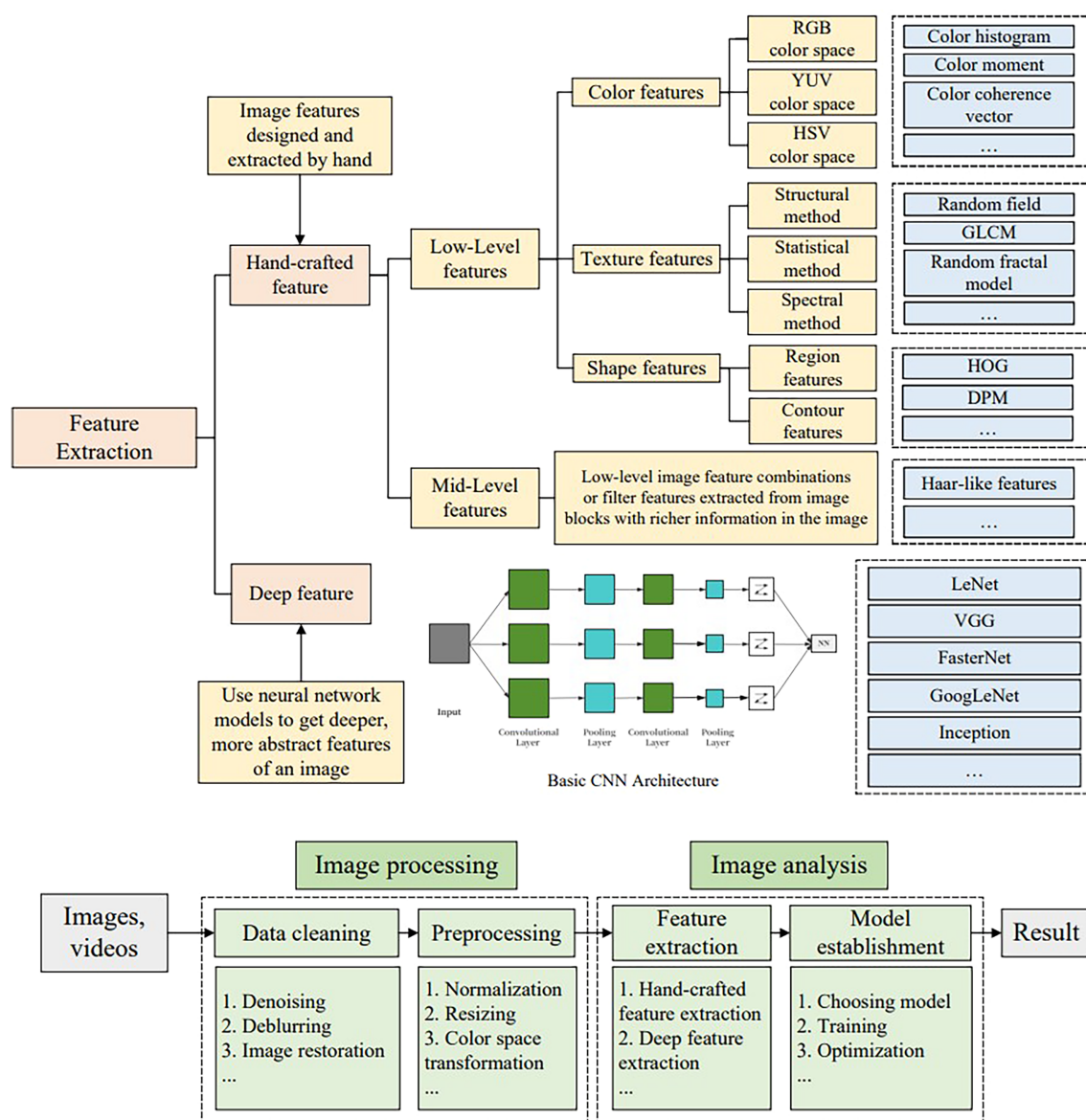


FIGURE 3 Feature extraction and computer vision task flow. CNN, convolutional neural network; DPM, deformable parts model; GLCM, gray-level co-occurrence matrix; HOG, histogram of oriented gradient; HSV, hue, saturation, value; RGB, red, green, blue; VGG, visual geometry group; YUV, luminance and chrominance.

texture, and color histograms, through multi-core learning to improve food image recognition (Hoashi et al., 2010; Taichi & Keiji, 2009). As the remarkable expressive power of convolutional neural networks becomes apparent, there are more and more works on food images based on deep features, such as using ResNet networks to directly extract visual features for dish image recognition (Ming et al., 2018), and extracting visual features from Network-In-Networks deep networks (Tanno et al., 2016). Presently, as deep learning continues to advance, deep features have demonstrated increasingly superior performance.

2.4 | Expert system

Expert system (or knowledge-based system) refers to the use of expert system techniques and control theory to design control systems that automate some of the tasks currently performed by human experts that traditional control systems are unable to perform (Linkens & Chen, 1995). The use of expert systems makes it possible to use the experience and knowledge accumulated over the years by experts in a particular field to solve problems that can only be solved by experts.

The general structure of an expert system consists of six parts: a human-computer interface, a reasoning machine, a knowledge base, a database, a knowledge acquisition organization, and an interpretation organization (Wanliang, 2011). Figure 4 shows the general structure of an expert system.

The core components of the expert system are the knowledge base and the reasoning engine, in the working process, the expert system according to the knowledge in the knowledge base and the facts provided by the user reasoning, constantly from the known facts to the unknown conclusions (intermediate results), and intermediate results to the database, as the new known facts for reasoning, to solve the problem from the unknown state to a known state.

Since the successful development of the first expert system DEN-DRAL in 1986, the development of expert system technology has been very rapid and has been applied to many fields such as mathematics, physics, chemistry, medicine, geology, meteorology, agriculture, law, education, transportation, machinery, art, etc., and even penetrated the major decision-making departments such as politics, economics, military, etc., which has brought great economic and

social benefits. However, due to the lack of expensive expert data and other reasons, the development of expert systems has been slow in recent years.

3 | APPLICATION OF AI TECHNOLOGY IN MEAT PROCESSING

3.1 | Carcass classification

Carcasses, the bodies of animals that have been slaughtered, eviscerated and otherwise processed, are essential raw materials for meat processing. Evaluation of carcass characteristics is the key to determining the value of slaughtered animals and grading carcasses through evaluation is an important step for meat processing companies to obtain high-quality raw materials for processing, but the systems used in different countries are different, some systems are based on yield prediction, while others are based on quality characteristics, Accurate detection and prediction of both is an important tool for the meat processing industry to minimize waste in the value chain while being able to respond to market demand (Gardner et al., 2021).

In most European countries, carcass grading is carried out by experts who rely on visual comparisons of individual carcass characteristics against criteria defined in each country, making carcass grading highly dependent on the knowledge, skills, and environment of the expert and not objective (De La Iglesia et al., 2020). Therefore, we need AI techniques to better perform the tasks of carcass classification, carcass quality detection, and yield prediction.

The following discussion is divided into two aspects carcass yield prediction and quality detection.

3.1.1 | Carcass production forecasts

Carcass yield, an important production criterion, is the final carcass weight as a percentage of the live weight of an animal after it has been stunned, bled, skinned, and eviscerated (Casey et al., 2003). Many factors affect carcass yield, such as animal breed, body fat distribution, slaughter treatment, etc., and how to improve the accuracy of predicting carcass yield is an important direction of research.

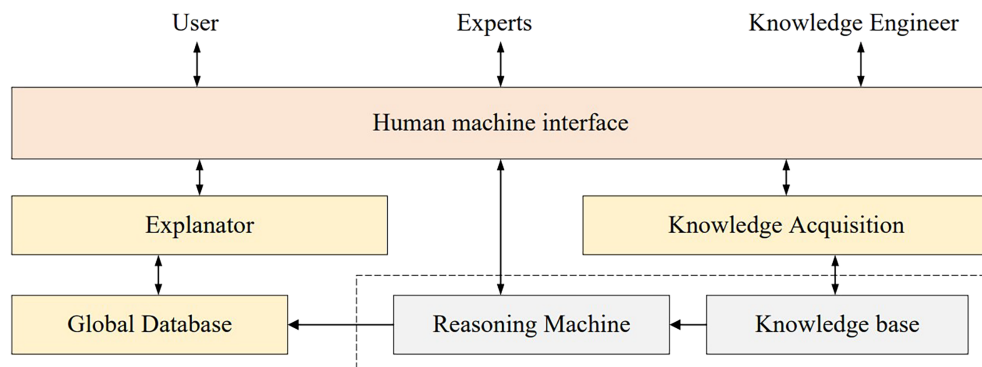


FIGURE 4 General structure of the expert system.

Machine-learning algorithms are widely used in carcass yield prediction. Shahinfar et al. (2018) used and compared nine different machine-learning algorithms and successfully applied the machine-learning methods in predicting carcass traits in sheep and found that the Random Forest Algorithm is the best method for predicting carcass traits. Gonçalves et al. (2021) proposed two image segmentation methods based on convolutional neural networks, Superpixel+CNN and SegNet methods, whose 97% pixel accuracy demonstrated the effectiveness and robustness of the methods in carcass segmentation of a variety of animals, contributing to the subsequent improvement in the accuracy of carcass grading. In addition, deep convolutional neural networks have been applied to ultrasound images to estimate intramuscular fat in pigs (Kvam & Kongsro, 2017). Deep learning algorithms have been applied to carcass images to predict beef carcass cut yield with comparable results to ML methods (Matthews et al., 2022). Supervised learning methods have also been applied to three-dimensional (3D) images to predict percent lean carcass yield in cattle (Alempijevic et al., 2021).

Another widely used technique in carcass yield prediction is CVS. Video image analysis (VIA) is one of the most researched techniques and solutions in beef, lamb, and pork carcass evaluation and its usefulness has been proven (Pabiou et al., 2011). Applications such as the prediction of lamb carcass composition, weight, and yield using VIA systems have demonstrated the great advantages of VIA technology in carcass yield prediction (Batista et al., 2021). In addition, Segura et al. (2021) found that CVS technology can accurately predict tissue composition in mature cows and Segura et al. (2023) evaluated the feasibility of two CVS systems in predicting beef carcass yield, and the results showed that CVS could predict post-processing weights, tissue compositions, and yield percentages.

Machine-learning algorithms and CVS techniques are often combined to achieve carcass classification. Chandraratne et al. (2007) extracted texture features from lamb chop images using three CVS feature extraction techniques and used artificial neural networks (ANNs) to classify lamb carcasses and compared the classification performance of ANNs with statistical discriminant function analysis (DFA) methods. The results clearly show that ANN is superior to DFA.

3.1.2 | Carcass quality assessment

Carcass quality assessment refers to the evaluation of factors in the carcass after slaughter that may affect the future visual and sensory characteristics of the meat product, such as fat color and tenderness (Delgado-Pando et al., 2021). A sound and accurate assessment of carcass quality is important to enable meat processors to respond to consumer demands.

CVS technology can be utilized for measuring the color of meat. In contrast to colorimetry and image analysis methods, which are restricted in assessing meat color because of light reflection and uneven meat surface color, CVS systems are capable of replicating a more authentic meat color. (Girolami et al., 2013).

CVS techniques are frequently employed in detecting disease states in carcasses. According to Chowdhury and Morey (2020), spectral imaging for poultry detection tends to be based on machine vision techniques. Detection of disease in carcasses of broilers, birds, and other animals can be done using CVS techniques and neural network models, as opposed to identifying healthy carcasses (Chao et al., 2007; Yang et al., 2005; Yang et al., 2009). Maes et al. (2023) stated that AI-based systems capable of automating the scoring of pleurisy pictures in pork carcasses, after slaughter, are being developed and require further research to train neural networks to recognize and score pneumonia. Spectroscopic techniques combined with CVS can be employed to investigate carcass surface contamination as an alternative approach. Kang et al. (2016) have devised an online detection and processing system for detecting surface contaminants on chicken carcasses based on multispectral imaging with a detection accuracy of 91.6%. Similarly, Wu et al. (2017) have succeeded in detecting blood, bile, and fecal matter diluted in the cecum and duodenum of chicken carcasses with a high recognition rate of 95.03% at the regional level.

3.1.3 | Strengths and weaknesses

AI techniques have become a crucial tool for evaluating and categorizing carcasses. Machine-learning and CVS techniques are particularly popular, demonstrating great value in predicting yield and assessing quality.

Compared to conventional approaches, AI-based techniques offer several benefits. First, they enable automated prediction of carcass yield and detection of surface contamination, resulting in reduced labor costs and improved detection efficiency. Second, AI technology is a non-invasive, prompt, and hygienic method of automatic assessment and measurement with high accuracy and reliability. This technology promotes food quality and safety.

However, there are certain limitations in the use of AI technology for assessing carcasses. Carcass inspection relying on optical technology and machine vision must take into account the differences caused by animals growing in different environments. Additionally, it must consider various environmental factors present in the processing plant, posing a challenge to the technology's effective implementation on the ground. One approach is to gather extensive data and construct a flexible classification system capable of adapting to diverse environments and circumstances (Chowdhury & Morey, 2020).

3.2 | Automation and intellectualization processing

3.2.1 | Analysis of the current situation

The Organisation for Economic Co-operation Development and the Food and Agricultural Organization (OECD-FAO) Agricultural Outlook 2023–2032 predicts that the global per capita demand for meat will

rise by 2% during the forecast period, from the base period of 2020–2022 to 2032. To meet the slow-growth demand, the meat supply will continue to expand, and global meat production will reach 382 million tons (+12%) by 2032 (OECD, Food and Agriculture Organization of the United Nations, 2023). Meat processing establishments, including abattoirs and meat processing factories, frequently experience a scarcity of skilled labor and substandard working conditions. Therefore, automated and intelligent meat processing lines are a crucial transition. Nevertheless, the degree of automation and intelligence implementation varies significantly across different processing segments since each segment necessitates distinct types of automated and intelligent processing.

A high degree of automation in meat processing has been realized for processing tasks that are continuously repetitive.

For instance, the initial phases of meat processing on the slaughter line have undergone substantial automation. The system of equipment utilized for meat cutting offers a distinctive illustration of the advanced automation in contemporary meat processing factories. The system comprises of two components: a laser scanner producing a 3D image of the cut, and a computer-guided high-speed rotary blade performing the cut, merging machine vision with computer science. It has the capability of carrying out 700 cuts per minute with a remarkable accuracy of $\pm 0.5\%$ that surpasses manual work, enhancing precision, speed, and hygiene levels. Meanwhile, meat processing systems, including the beef carcass cutting system, which combines sensors to produce precise carcass images with computer-controlled cutting, and the water jet knife automatic skin trimming system for fish fillets, where a computer algorithm ascertains the optimum trimming pattern to reduce waste, demonstrate the advanced level of automation and intelligence in the primary processing section of meat processing plants (Barbut, 2020).

Meat processing automation and intelligence have been extensively utilized in primary processing. According to Mason et al. (2022), technological advancements such as optical technology and AI (AI) can enhance the speed and durability of cutting tools, providing decision-making and data analysis assistance, which is intriguing for smart knives. Various technologies for developing future robotic meat-cutting smart knives were examined by the authors. Esper et al. (2021) summarized the automated systems concerning slaughterhouses and discussed various meat processing systems, including Frontmatec AiRA, Mayekawa Hamdan-RX, and other systems that involve different primary meat processing aspects, such as cutting and boning. These processing systems incorporate different vision technologies and have varying degrees of automation. However, they all provide innovative and intelligent systems to slaughterhouses, enabling new opportunities for meat processors.

Additionally, automated and intelligent meat processing finds widespread use in other areas of meat processing where tasks tend to be repetitive. For instance, to decrease the monotony of unloading crates manually and catching conscious birds in them to place them on a moving yoke line, semi-automated or fully automated control atmosphere stunning systems have been implemented, which have

shown efficacy in accelerating poultry production lines (Barbut, 2014). Electrical stimulation technology improves the texture and mouthfeel of meat by using electrical currents to stimulate the muscle tissue after slaughter. Simmons et al. (2006) developed a system that allows customization of the stimulation parameters (number, intensity, and duration) to the needs of a specific beef or lamb carcass while the carcass is moving through the production line and is capable of predicting the final pH and pressure-pain of the meat, which not only improves the product's consistency but also customize quality attributes to specific market requirements. The automated food factory at Zhengda in Qinhuangdao, Hebei, China employs AI technology to create a clever dumpling production line that is completely automated. From mixing to stirring and kneading, this process generates up to 100,000 dumplings per hour, which considerably enhances the economic efficiency of the company (Xiaoshan & Weiwei, 2021).

AI technology has been implemented in every part of the meat processing industry due to its ability to simplify production, increase efficiency, and enhance process technology. The use of AI technology for continuous and repeated operations can boost the automation level of meat processing and achieve intelligent processing. It can achieve machine learning by using AI algorithms, improving the accuracy of processing and production continually, and performing a significant function in enhancing efficiency and reducing expenses, while providing more intelligent and efficient production techniques for the meat processing sector.

However, in the need for greater flexibility in the processing chain, meat processing automation, intelligence is still relatively low. More complete meat processing automation and intelligent realization are limited by several practical conditions. On the one hand, meat processing involves multiple links and complex operations. The processing environment may also have noise, dust, and other disturbing factors, that affect the sensors and equipment. In addition, there may be variations in the characteristics and quality of meat, such as size and shape. These external factors and the variability of the meat itself bring some difficulties to the realization of automation and intelligence. On the other hand, the realization of automation and intelligence in meat processing requires the investment of large amounts of capital and technological resources. In addition to the development and application of algorithms, it is necessary to be equipped with advanced sensors, equipment, and systems for data acquisition, processing, and control. The high cost of these devices and the need for professionals to maintain and operate them increases the cost and difficulty of implementation, and it is difficult to carry out the application of automation and intelligent meat processing for smaller-scale meat processing plants. Through the development of software algorithms and hardware research, it can greatly reduce interference and reduce the cost of sensors and other hardware equipment. With the continuous progress and innovation of technology, we believe that these challenges will be gradually overcome and bring more efficient and intelligent production methods to the meat processing industry.

3.2.2 | Trends in development

To more complete meat processing automation, intelligent development is the inevitable trend after the iteration of meat processing technology, the development of many directions, and the development prospects. Human-machine collaboration, Meat Factory Cells (MFCs), and the introduction of more AI algorithms are all important development directions for future meat processing automation and intelligence.

Human-robot collaboration is one of the important development directions for future meat processing automation and intelligence. Green et al. (2008) pointed out that human-robot collaboration utilizes both human problem-solving ability and the speed and flexibility of the robot. It allows the robot to work autonomously while retaining the ability to ask for help when guidance is needed. As a major meat change, human-robot collaboration is increasingly seen as a key feature of future factory automation (Demir et al., 2019). However, unlike the widespread use of human-robot collaboration in other production lines, such as automotive production lines, human-robot collaboration is still less used and less commercialized in meat processing.

The MFC is also an important future direction for the automation and intelligence of meat processing. Unlike the current popularity of processing using production lines with many specialized stations, MFC allows carcasses to be processed entirely in the same location, in a concept that replaces short and repetitive line operations with robust, flexible, and scalable automated and semi-automated processing in the processing cell. Advances in AI technology have made it possible for processing systems to process and respond to real-time inputs when faced with tasks in which the target material varies due to biological and processing factors (e.g., shape, position, color change). This has also greatly accelerated the popularization and development of the MFC concept (Mason et al., 2021).

The introduction of more AI algorithms is of great significance to realize the automation and intelligence of meat processing. Nowadays, although we can see many software programs and special algorithms in the processing field, AI algorithms, as an emerging technology with self-learning, self-adaptation, self-organization, and other characteristics, have not yet been widely used in the field of meat processing intelligence. Unlike traditional software programs and specialized algorithms, AI algorithms can continuously learn and adapt to the existence of various uncertainties in meat processing, such as changes in the shape, size, and quality of meat, the existence of complex production processes and multiple links, and even the existence of changes in the processing environment, to reduce the need for human intervention and improve the overall efficiency and productivity.

3.3 | Meat-quality inspection

3.3.1 | Overview of meat-quality inspection applications

Meat-quality inspection is the process of evaluating the quality of meat, which usually involves several quality indicators, including color,

tenderness, fat content, meat texture, etc. It can evaluate the freshness, taste, and nutritional value of meat, to ensure the quality of meat and the safety of consumption, and to bring a better experience for consumers.

AI technology is widely used in meat-quality inspection, often combined with spectral technology such as multispectral imaging, hyperspectral imaging, and near-infrared spectral imaging, chromatographic technology such as ultra-fast gas chromatography, electronic nose, electronic tongue, and other sensor technologies to complete the grading and prediction of meat freshness and other qualities.

Currently, there are many studies devoted to applying AI techniques to meat-quality detection. These studies have achieved certain results and have shown great potential in improving detection efficiency and accuracy. Table 1 summarizes some representative existing research on AI techniques for meat-quality detection, presenting them in terms of research objects, AI technology applications, other methods, and research results.

AI techniques can also be used for meat-quality monitoring tasks in meat processing plants. For example, AI techniques such as expert systems have been included in the discussion of the processing monitoring process. Blagoveshchenskiy et al. (2020) discussed the shortcomings of the existing methods in product quality evaluation and proposed a solution for the monitoring and prediction of product quality indicators by using intelligent techniques and databases, developing a database of product quality parameter monitoring intelligent expert system. In addition, Yang et al. (2010) used radial basis function neural networks, fuzzy data processing techniques, and other techniques to develop a pork production and processing monitoring system capable of monitoring, acquiring, and exchanging pork production and processing data in real time.

3.3.2 | Challenges and perspectives

The above research has shown good performance in the field of meat-quality testing and has a very promising application, but still faces many problems and challenges.

On the one hand, how to obtain high-quality data is the first challenge encountered in meat-quality testing. Current research focuses on modeling and analysis of existing data, and when the data quality is not good, the model obtained from the research will be invalid to a certain extent. In practice, meat-quality data rely on acquiring a large amount of sensor data, but the quality and diversity of the data may be affected by many factors, such as the sensor accuracy, the data acquisition environment, and the differences in meat. Therefore, how to ensure data quality is an urgent problem.

On the other hand, further exploration and development are needed in the combination, optimization, and system integration of AI technologies. The current research on AI technology in meat-quality inspection is still mainly focused on single inspection tasks, and there are relatively few studies on real-time data acquisition and quality monitoring. We need to study how to combine the short-time single meat-quality inspection technology to form a comprehensive meat-

TABLE 1 Application of artificial intelligence technology in meat-quality inspection.

Research target	Used model(s)	Other method(s)	Results of the study	References
Sardine	FLM	–	$R = 0.97, 0.95$, and 1	Zare and Ghazali (2017)
Pork meat	ANN	UFGC and E-nose	Fresh: 80% Frozen-then-thawed: 85% spoiled meat: 90%	Górska-Horczyzak et al. (2017)
Beef fillet	ANFIS, AFLS, and MLP	E-nose	AFLS MSE: 0.1540 ANFIS MSE: 0.2214 MLP MSE: 0.2397	Kodogiannis and Alshejari (2016)
Beef	MLP, ANFIS, and AFINN	Multispectral images	ANFIS classification accuracy: 93.75%	Alshejari and Kodogiannis (2017)
Boneless pork	CSV and SVM	–	Measured pork color Score accuracy: 92.5% Measured pork marbling Score accuracy: 75.0%	Sun et al. (2018)
Parabramis pekinensis	BP-ANN and CVS	NIR spectra	BP-ANN prediction set performance: 93.33%	Huang et al. (2016)
Frozen salmon	CVS and TreeBagger classifier	HSI	Optimization model cross-validation CCR: 0.978	Xu et al. (2017)
Chicken meat	SVM and REPTree	NIR spectra	REPTree classification precision: 77.2%	Sylvio et al. (2018)
Fish	MLPNN	E-nose	MLPNN correct rate: 96.87%	Vajdi et al. (2019)
Lamb meat	SVM and MLP classifier	FTIR spectra	MLP-SVM: 91.80% MLP-PCA: 93.89%	Alaiz et al. (2020)
Pork meat	ANN	–	pH maximal error: ± 0.2 Water activity maximal error: ± 0.015	Ndob et al. (2017)

Abbreviations: AFINN, the adaptive fuzzy inference neural network; AFLS, adaptive fuzzy logic system; ANFIS, adaptive neuro-fuzzy inference system; ANN, artificial neural networks; BP-ANN, backpropagation artificial neural network; CCR, correct classification rate; CSV, comma-separated values; CVS, computer vision system; FLM, fuzzy logic model; FTIR, Fourier-transform infrared; HSI, hue, saturation, intensity; MLP, multi-layer perceptron; MLPNN, multi-layer perceptron neural network; MSE, mean square error; NIR, near infrared; PCA, principal component analysis; SVM, support vector machine; UFGC, ultra-fast gas chromatography.

quality inspection and monitoring system with multiple AI technologies, to improve the comprehensive performance of the whole system.

Future research can continue to explore more advanced AI techniques and combine domain expert knowledge to improve the accuracy and real-time performance of meat-quality detection. In conclusion, although we have achieved a series of research results in the field of meat-quality detection, further exploration and innovation are still needed to overcome the existing problems and challenges, and then realize more accurate, efficient, and real-time meat-quality detection, improve the monitoring and control of meat quality, and provide more reliable and efficient quality assurance for the meat industry.

4 | CONCLUSION

AI technology has great potential in meat processing to improve productivity, ensure product quality and safety, and optimize management. Through automation and intelligent processing, meat processors can achieve higher yields and speeds, and reduce labor costs and error rates. In addition, by utilizing AI technology for

carcass sorting and meat-quality inspection, they can improve inspection efficiency and provide more accurate quality control and quality inspection. However, the application of AI in meat processing still faces some challenges. On the one hand, the popularization of technology and cost issues need to be resolved, and the construction of intelligent production lines and systems requires a large investment in capital and cutting-edge technology. On the other hand, due to the existence of differences in environments and so on, the models obtained will have certain limitations, and it is difficult to generalize them to the same meat processing enterprises in different environments or different meat processing enterprises.

To summarize, the application of AI in meat processing has a broad prospect, but there are also problems and challenges, we believe that AI technology will play a greater role in the meat processing industry and make a sustained and outstanding contribution to the development of the industry.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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REFERENCES

- Adesokan, H., & Raji, A. O. Q. (2014). Safe meat-handling knowledge, attitudes and practices of private and government meat processing plants' workers: Implications for future policy. *Journal of Preventive Medicine and Hygiene*, 55, 10–16.
- Alaiz, R., & Parnell, A. (2020). A machine learning approach for lamb meat quality assessment using FTIR spectra. *IEEE Access*, 8, 52385–52394. <https://doi.org/10.1109/ACCESS.2020.2974623>
- Alimpijevic, A., Vidal-Calleja, T., Falque, R., Quin, P., Toohey, E., Walmsley, B., & McPhee, M. (2021). Lean meat yield estimation using a prototype 3D imaging approach. *Meat Science*, 181, 108470. <https://doi.org/10.1016/j.meatsci.2021.108470>
- Alshejari, A., & Kodogiannis, V. S. (2017). An intelligent decision support system for the detection of meat spoilage using multispectral images. *Neural Computing and Applications*, 28(12), 3903–3920. <https://doi.org/10.1007/s00521-016-2296-6>
- Barbut, S. (2014). Review: Automation and meat quality-global challenges. *Meat Science*, 96(1), 335–345. <https://doi.org/10.1016/j.meatsci.2013.07.002>
- Barbut, S. (2020). Meat Industry 4.0: A distant future? *Animal Frontiers*, 10(4), 38–47. <https://doi.org/10.1093/af/vfaa038>
- Batista, A. C., Santos, V., Afonso, J., Guedes, C., Azevedo, J., Teixeira, A., & Silva, S. (2021). Evaluation of an image analysis approach to predicting primal cuts and lean in light lamb carcasses. *Animals*, 11(5), 1368.
- Blagoveshchenskiy, I. G., Blagoveshchenskiy, V. G., Besfamilnaya, E. M., & Sumerin, V. A. (2020). Development of databases of intelligent expert systems for automatic control of product quality indicators. *Journal of Physics: Conference Series*, 1705(1), 012019. <https://doi.org/10.1088/1742-6596/1705/1/012019>
- Casey, N. H., Niekerk, W. A., & Webb, E. C. (2003). Goat meat. In B. Caballero (Ed.), *Encyclopedia of food sciences and nutrition* (2nd ed., pp. 2937–2944). Academic Press.
- Chandraratne, M. R., Kulasiri, D., & Samarasinghe, S. (2007). Classification of lamb carcass using machine vision: Comparison of statistical and neural network analyses. *Journal of Food Engineering*, 82(1), 26–34. <https://doi.org/10.1016/j.jfoodeng.2007.01.003>
- Chao, K., Yang, C. C., Chen, Y. R., Kim, M. S., & Chan, D. E. (2007). Hyper-spectral-multispectral line-scan imaging system for automated poultry carcass inspection applications for food safety. *Poultry Science*, 86(11), 2450–2460. <https://doi.org/10.3382/ps.2006-00467>
- Chowdhury, E. U., & Morey, A. (2020). Application of optical technologies in the US poultry slaughter facilities for the detection of poultry carcass condemnation. *British Poultry Science*, 61(6), 646–652. <https://doi.org/10.1080/00071668.2020.1792833>
- De La Iglesia, D. H., González, G. V., García, M. V., Rivero, A. J. L., & De Paz, J. F. (2020). Non-invasive automatic beef carcass classification based on sensor network and image analysis. *Future Generation Computer Systems*, 113, 318–328. <https://doi.org/10.1016/j.future.2020.06.055>
- Delgado-Pando, G., Allen, P., Troy, D. J., & McDonnell, C. K. (2021). Objective carcass measurement technologies: Latest developments and future trends. *Trends in Food Science & Technology*, 111, 771–782.
- Demir, K. A., Döven, G., & Sezen, B. (2019). Industry 5.0 and human-robot co-working. *Procedia Computer Science*, 158, 688–695. <https://doi.org/10.1016/j.procs.2019.09.104>
- Esper, I. D. M., From, P. J., & Mason, A. (2021). Robotisation and intelligent systems in abattoirs. *Trends in Food Science & Technology*, 108, 214–222.
- Gardner, G. E., Apps, R., McColl, R., & Craigie, C. R. (2021). Objective measurement technologies for transforming the Australian & New Zealand livestock industries. *Meat Science*, 179, 108556.
- Girolami, A., Napolitano, F., Faraone, D., & Braghieri, A. (2013). Measurement of meat color using a computer vision system. *Meat Science*, 93(1), 111–118. <https://doi.org/10.1016/j.meatsci.2012.08.010>
- Gonçalves, D. N., Weber, V. A. d. M., Pistori, J. G. B., Gomes, R. d. C., de Araujo, A. V., Pereira, M. F., Gonçalves, W. N., & Pistori, H. (2021). Carcass image segmentation using CNN-based methods. *Information Processing in Agriculture*, 8(4), 560–572. <https://doi.org/10.1016/j.inpa.2020.11.004>
- Górska-Horczyzak, E., Horczyzak, M., Guzek, D., Wojtasik-Kalinowska, I., & Wierzbicka, A. (2017). Chromatographic fingerprints supported by artificial neural network for differentiation of fresh and frozen pork. *Food Control*, 73, 237–244.
- Green, S. A., Billingham, M., Chen, X., & Chase, J. G. (2008). Human-robot collaboration: A literature review and augmented reality approach in design. *International Journal of Advanced Robotic Systems*, 5(1), 1. <https://doi.org/10.5772/5664>
- Grujić, R. D. (2010). *Meat in human nutrition*. Quality of Life (Banja Luka).
- Hoashi, H., Joutou, T., & Yanai, K. (2010). Image recognition of 85 food categories by feature fusion. Paper presented at the 2010 IEEE International Symposium on Multimedia, 13–15 December 2010. Quality of Life (Banja Luka).
- Huang, X., Xu, H., Wu, L., Dai, H., Yao, L., & Han, F. (2016). A data fusion detection method for fish freshness based on computer vision and near-infrared spectroscopy. *Analytical Methods*, 8(14), 2929–2935. <https://doi.org/10.1039/C5AY03005F>
- Kang, R., Yang, K., Xia Zhang, X., Wu, W., & Jie Chen, K. (2016). Development of online detection and processing system for contaminants on chicken carcass surface. *Applied Engineering in Agriculture*, 32(1), 133–139. <https://doi.org/10.13031/aea.32.11200>
- Kodogiannis, V. S., & Alshejari, A. (2016). Neuro-fuzzy based identification of meat spoilage using an electronic nose. Paper presented at the 2016 IEEE 8th International Conference on Intelligent Systems (IS), 4–6 September 2016.
- Kruger, N., Janssen, P., Kalkan, S., Lappe, M., Leonardi, A., Piater, J., Rodriguez-Sanchez, A. J., & Wiskott, L. (2013). Deep hierarchies in the primate visual cortex: What can we learn for computer vision? *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(8), 1847–1871. <https://doi.org/10.1109/TPAMI.2012.272>
- Kvam, J., & Kongsro, J. (2017). In vivo prediction of intramuscular fat using ultrasound and deep learning. *Computers and Electronics in Agriculture*, 142, 521–523. <https://doi.org/10.1016/j.compag.2017.11.020>
- Linkens, D. A., & Chen, M. (1995). Expert control systems—I. Concepts, characteristics and issues. *Engineering Applications of Artificial Intelligence*, 8(4), 413–421.
- Liu, P. R., Lu, L., Zhang, J. Y., Huo, T. T., Liu, S. X., & Ye, Z. W. (2021). Application of artificial intelligence in medicine: An overview. *Current Medical Science*, 41(6), 1105–1115. <https://doi.org/10.1007/s11596-021-2474-3>
- Maes, D., Sibila, M., Pieters, M., Haesebrouck, F., Segalés, J., & de Oliveira, L. G. (2023). Review on the methodology to assess respiratory tract lesions in pigs and their production impact. *Veterinary Research*, 54(1), 8. <https://doi.org/10.1186/s13567-023-01136-2>
- Mason, A., Esper, I., Korostynska, O., Haidegger, T., Popov, A., Christensen, L. B., & Alvseike, O. (2021). *The Meat Factory Cell: A new way of thinking for meat producers*. Paper Presented at the 2021 IEEE

- 21st International Symposium on Computational Intelligence and Informatics (CINTI), 18–20 November 2021.
- Mason, A., Romanov, D., Cordova-Lopez, L. E., Ross, S., & Korostynska, O. (2022). Smart knife: Technological advances towards smart cutting tools in meat industry automation. *Sensor Review*, 42(1), 155–163. <https://doi.org/10.1108/SR-09-2021-0315>
- Matthews, D., Pabiou, T., Evans, R. D., Beder, C., & Daly, A. (2022). Predicting carcass cut yields in cattle from digital images using artificial intelligence. *Meat Science*, 184, 108671. <https://doi.org/10.1016/j.meatsci.2021.108671>
- Medathati, N. V. K., Neumann, H., Masson, G. S., & Kornprobst, P. (2016). Bio-inspired computer vision: Towards a synergistic approach of artificial and biological vision. *Computer Vision and Image Understanding*, 150, 1–30.
- Ming, Z. Y., Chen, J., Cao, Y., Forde, C., Ngo, C. W., & Chua, T. S. (2018). Food Photo Recognition for Dietary Tracking: System and Experiment. In: K. Schoeffmann (Ed.), *MultiMedia modeling*. MMM 2018. Lecture Notes in Computer Science, Vol. 10705. Springer. https://doi.org/10.1007/978-3-319-73600-6_12
- Mulvenna, M. D., Woodham, C., & Gregg, J. B. (1991). Artificial intelligence applications in geology: A case study with explorer. Paper Presented at the AI and Cognitive Science '90, London.
- Ndob, A. M., & Lebert, A. (2017). Prediction of pH and aw of pork meat by a thermodynamic model: New developments. *Meat Science*, 138, 59–67.
- OECD, Food and Agriculture Organization of the United Nations. (2023). *OECD-FAO Agricultural Outlook 2023-2032*. OECD Publishing.
- Pabiou, T., Fikse, W. F., Cromie, A. R., Keane, M. G., Näsholm, A., & Berry, D. P. (2011). Use of digital images to predict carcass cut yields in cattle. *Livestock Science*, 137(1), 130–140. <https://doi.org/10.1016/j.livsci.2010.10.012>
- Raj, A., Jayaram, R., Dev, S., & Ravikumar, S. (2013). A review on recent advances in meat processing. *International Journal of Current Microbiology and Applied Biosciences*, 2, 169–176.
- Russell, S., & Norvig, P. (2004). *Artificial intelligence—A modern approach*. Tsinghua University Press.
- Seaton, M. (2022). Lessons in automation of meat processing. *Animal Frontiers*, 12(2), 25–31. <https://doi.org/10.1093/af/vfac022>
- Segura, J., Aalhus, J. L., Prieto, N., Larsen, I. L., Juárez, M., & López-Campos, Ó. (2021). Carcass and primal composition predictions using camera vision systems (CVS) and dual-energy X-ray absorptiometry (DXA). *Technologies on Mature Cows*, 10(5), 1118.
- Segura, J., Aalhus, J. L., Prieto, N., Zawadski, S., Scott, H., & López-Campos, Ó. (2023). Prediction of primal and retail cut weights, tissue composition and yields of youthful cattle carcasses using computer vision systems; whole carcass camera and/or ribeye camera. *Meat Science*, 199, 109120. <https://doi.org/10.1016/j.meatsci.2023.109120>
- Shahinfar, S., Kelman, K., & Kahn, L. (2018). Prediction of sheep carcass traits from early-life records using machine learning. *Computers and Electronics in Agriculture*, 156, 159–177.
- Simmons, N. J., Daly, C. C., Mudford, C. R., Richards, I., Jarvis, G., & Pleiter, H. (2006). Integrated technologies to enhance meat quality—An Australasian perspective. *Meat Science*, 74(1), 172–179. <https://doi.org/10.1016/j.meatsci.2006.05.007>
- Sun, X., Young, J., Liu, J. H., & Newman, D. (2018). Prediction of pork loin quality using online computer vision system and artificial intelligence model. *Meat Science*, 140, 72–77. <https://doi.org/10.1016/j.meatsci.2018.03.005>
- Sylvio, B., Da Costa Barbon, A. P. A., Mantovani, R. G., & Barbin, D. F. (2018). Machine learning applied to near-infrared spectra for chicken meat classification. *Journal of Spectroscopy*, 2018, 8949741.
- Taichi, J., & Keiji, Y. (2009). A food image recognition system with multiple kernel learning. Paper Presented at the 2009 16th IEEE International Conference on Image Processing (ICIP), 7–10 November 2009.
- Tanno, R., Okamoto, K., & Yanai, K. (2016). *DeepFoodCam: A DCNN-based real-time mobile food recognition system*. Proceedings of the 2nd International Workshop on Multimedia Assisted Dietary Management.
- Vajdi, M., Varidi, M. J., Varidi, M., & Mohebbi, M. (2019). Using electronic nose to recognize fish spoilage with an optimum classifier. *Journal of Food Measurement and Characterization*, 13, 1205–1217.
- Wang, H. H. (2022). The perspective of meat and meat-alternative consumption in China. *Meat Science*, 194, 108982. <https://doi.org/10.1016/j.meatsci.2022.108982>
- Wanliang, W. (2011). *Introduction to artificial intelligence*. China Higher Education Press.
- Wu, W., Chen, G. Y., Wu, M. Q., Yu, Z. W., & Chen, K. J. (2017). Detection of diluted contaminants on chicken carcasses using a two-dimensional scatter plot based on a two-dimensional hyperspectral correlation spectrum. *Applied Optics*, 56(9), D72–D78. <https://doi.org/10.1364/ao.56.000d72>
- Xiaoshan, Z., & Weiwei, L. (2021). Research on the application of artificial intelligence technology in food processing. *Meat Research*, 35(2), 10034.
- Xu, J. L., & Sun, D. W. J. I. J. o. R.-r. I. D. F. (2017). Identification of freezer burn on frozen salmon surface using hyperspectral imaging and computer vision combined with machine learning algorithm. *International Journal of Refrigeration*, 74, 151–164.
- Yang, C.-C., Chao, K., & Chen, Y.-R. (2005). Development of multispectral image processing algorithms for identification of wholesome, septicemic, and inflammatory process chickens. *Journal of Food Engineering*, 69(2), 225–234. <https://doi.org/10.1016/j.jfoodeng.2004.07.021>
- Yang, C.-C., Chao, K., & Kim, M. S. (2009). Machine vision system for online inspection of freshly slaughtered chickens. *Sensing and Instrumentation for Food Quality and Safety*, 3(1), 70–80. <https://doi.org/10.1007/s11694-008-9067-8>
- Yang, Y., Zhang, K., & Chai, Y. (2010). *An implementation of intelligent monitoring system for food processing*. Paper Presented at the 2010 8th World Congress on Intelligent Control and Automation, 7–9 July 2010.
- Zare, D., & Ghazali, H. M. (2017). Assessing the quality of sardine based on biogenic amines using a fuzzy logic model. *Food Chemistry*, 221, 936–943. <https://doi.org/10.1016/j.foodchem.2016.11.071>
- Zhu, F., Bosch, M., Schap, T., Khanna, N., Ebert, D. S., Boushey, C. J., & Delp, E. J. (2011). Segmentation assisted food classification for dietary assessment. *Proceedings of SPIE the International Society for Optical Engineering*, 7873, 78730b. <https://doi.org/10.1117/12.877036>

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